

강의계획서

2026학년도 1학기

과목명	알고리즘		교과목 번호	MC4206	학점	3	강의실	창708
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1. 교과목 개요

Course Objectives

On completion of this course, you'll hopefully be able to:

- Model/analyze real-world problems and design algorithmic solutions along with mathematical proofs using discrete structures and combinatorial methods.
- Understand/appreciate limits of computation.

Contents

- Intractability (algorithm design anti-patterns)
 - NP-completeness, Karp reduction vs. Cook/Turing reduction, co-NP
 - PSPACE-completeness, #P-completeness
- Algorithm design/analysis patterns for tractable problems
 - Greedy strategy (matroids), divide-and-conquer (e.g. FFT)
 - Dynamic programming (convex hull trick, D&C opt., Knuth opt.)
 - Polynomial-time reduction (e.g. max-flow/min-cut, 2-SAT)
 - Randomization (ZPP, RP, BPP)
 - Amortized analysis (e.g. disjoint-set data structure)
- Algorithm design patterns for intractable problems
 - Subproblem analysis (e.g. fixed-parameter tractability)
 - Approximation (e.g. pricing, LP relaxation, rounding/trimming for FPTAS)
- Algorithms for online problems
- Quantum algorithms and quantum-resistant cryptography
 - Quantum computation, BQP
 - Shor's algorithms for discrete logarithm and factoring
 - Lattice-based public-key encryption schemes (learning with errors)
- Generic algorithms for mathematical programming
 - Simplex method for LP, duality, ellipsoid method (separation oracle)
 - Interior point method, gradient descent for convex optimization
 - Cutting-plane method, branch-and-cut method for integer linear programming

2. 중점 목표 역량

교과 핵심 역량			
problem-solving skills	logical thinking	problem discovery and definition	creativity

3. 교재 및 참고도서

가. 주교재 : 자체 제작 수업 자료

나. 참고 서적

- “Algorithm Design”, J. Kleinberg and E. Tardos
- “Computers and Intractability: A Guide to the Theory of NP-Completeness”, M. Garey et al.
- “Introduction to Theory of Computation”, M. Sipser
- “Introduction to Algorithms”, T. Cormen, C. Leiserson, R. Rivest, and C. Stein
- “Algorithms”, S. Dasgupta, C. H. Papadimitriou, and U. V. Vazirani
- “Introduction to Linear Optimization”, D. Bertsimas and J. Tsitsiklis
- “Linear Programming: Foundations and Extensions”, R. J. Vanderbei
- “Discrete Optimization”, G. Parker and R. Rardin
- “Convex Optimization”, S. Boyd and L. Vandenberghe
- “Approximation Algorithms”, U. V. Vazirani
- “Online Computation and Competitive Analysis”, A. Borodin and R. El-Yaniv
- “Computational Complexity”, S. Arora and B. Barak
- “An Intensive Introduction to Cryptography”, B. Barak
- “A Decade of Lattice Cryptography”, C. Peikert
- “Combinatorial Optimization: Polyhedra and Efficiency”, A. Schrijver
- “Discrete and Combinatorial Mathematics”, R. Grimaldi

4. 수업 방법 :

가. 강의

5. 평가 기준 및 방법

가. 평가 기준

과제 (15회)	기말고사	출결 및 태도	합계
10%	80%	10%	100%

나. 평가 방법

- 15 problem sets
 - Scoring policy: full score if 2/3 submitted, otherwise 0 point
- Final exam: 9:00-23:00 on the Saturday right before the final exam period
- Make-up final exam for officially approved absence (e.g. COVID, illness, out-of-school competition, conference)
 - immediately after the day all students are released from quarantine/...
 - 0 point if do not show up on the make-up final test
- Late-work policy: not accepted

6. 학점 부여 기준

- 출결 및 태도: 총 수업 시간의 3/4 이상 부여시 100% 부여
- 병결/공결 등으로 정규 기말고사에 미응시 경우: 대체시험 실시하여 그대로 기말고사 점수로 반영
- 과목우수상 기준: 원점수 기준으로 정렬하여 결정

7. 주별 강의 계획

주(기간)	주제 및 내용	(비고)
제1주 2.23.~3.1.	1. Polynomial-time reductions #1 • Algorithmic problems: decision vs. search vs. optimization • Polynomial-time reductions • Standard reduction techniques (simple equivalence, restriction, local replacement, component design) 2. Polynomial-time reductions #2 • 3-SAT vs. 2-SAT • Karp reduction vs. Cook/Turing reduction • Self-reducibility • Anatomy of poly-time reductions: a certifier-based perspective 3. Reduction examples #1 • $3\text{-SAT} \leq_p 3\text{-dimensional matching}$ • $3\text{-SAT} \leq_p \text{directed Hamiltonian cycle} \leq_p \text{Hamiltonian cycle} \leq_p \text{TSP}$ • $3\text{-SAT} \leq_p \text{longest simple path}$	
핵심역량	problem-solving skills, logical thinking	
제2주 3.2.~3.8.	4. Reduction examples #2 • $3\text{-SAT} \leq_p 3\text{-colorability} \leq_p 3\text{-colorability on planar graphs}$ • Planar k-colorability • $3\text{-SAT} \leq_p \text{subset sum}$ • $3\text{-DM} \leq_p \text{subset sum}$ • $\text{Subset sum} \leq_p \text{partition}$ • $\text{Subset sum} \leq_p \text{scheduling with release time}$ 5. NP-completeness • Certifier-based definition of the class NP • Cook-Levin theorem • NP-easy vs. NP-hard vs. NP-complete • How to cope with intractability • Complexity classes beyond NP (co-NP, PSPACE, ZPP, BPP, RP, #P, PPAD, BQP) 6. co-NP • Asymmetry of NP • Problems in $\text{NP} \cap \text{co-NP}$	
핵심역량	problem-solving skills, logical thinking	
제3주 3.9.~3.15.	7. PSPACE • Quantified satisfiability • Planning problems • PSPACE-completeness 8. 2-SAT • Kosaraju's algorithm for identifying strongly connected components (SCCs) • Implication graph for 2-CNF, algorithm for 2-SAT • Reductions to 2-SAT	
핵심역량	problem-solving skills, logical thinking	
제4주 3.16.~3.22.		

	9. Greedy algorithms for simple problems (exchange argument) • Selecting items to maximize total weights • Cashier's algorithm for coin changing • Truck driver's algorithm to select breakpoints 10. Greedy scheduling algorithms • Interval scheduling ("stays ahead" argument) • Interval partitioning (structural property of solution) • Scheduling to minimize lateness (exchange argument) • Farthest-in-future (FF) policy for offline caching 11. Huffman codes for data compression • Optimal prefix codes • Huffman encoding	
핵심역량	problem-solving skills, logical thinking	
제5주 3.23.~3.29.	12. Divide-and-conquer algorithms • Counting inversions using merge sort • Closest pair of points • Karatsuba's algorithm for integer multiplication • Strassen's algorithm for matrix multiplication 13. Fast Fourier Transform (FFT) • Convolutions & polynomial multiplications • Fast discrete Fourier transform • FFT circuits, quantum circuits for discrete Fourier transform	
핵심역량	problem-solving skills, logical thinking	
제6주 3.30.~4.5.	14. Dynamic programming pattern #1 • Binary choice: weighted interval scheduling • Multi-way choice: segmented least squares • Additional variable: knapsack (pseudo-polynomial time algorithm) 15. Dynamic programming pattern #2 • DP over intervals: RNA secondary structure, CYK parsing • Multi-dimensional: edit distance (linear space) 16. DP optimization techniques • Convex hull trick • Divide-and-conquer optimization • Knuth optimization	
핵심역량	problem-solving skills, logical thinking	
제7주 4.6.~4.12.	21. Amortized analysis • Aggregate analysis vs. accounting method vs. potential method • Disjoint-set data structure, union-find with path compression 17. Max-flow & min-cut • Weak duality of flows and cuts • Ford-Fulkerson method • Max-flow min-cut theorem • Flow Integrality • Capacity scaling method 18. Bipartite matching • Reduction from bipartite matching to max flow • Hall's theorem for perfect matching: max-flow/min-cut thm vs. Dilworth's thm 19. Reductions to max flow • Circulation with demands: reduction to max flow • Survey design: reduction to circulation with demands • Edge-disjoint paths: reduction to max flow • Network connectivity: duality with edge-disjoint paths 20. Reductions to min cut • Image segmentation: reduction to min cut • Project selection: reduction to min cut • Open pit mining: reduction to project selection	

핵심역량	problem-solving skills, logical thinking	
제8주 4.13.~4.19.	중간고사 (4.14~4.17.)	
주(기간)	주제 및 내용	(비고)
제9주 4.20.~4.26.	22. Simplex algorithm #1 • LP in standard form vs. slack form • Formulating discrete optimization problems as linear programs • Pivoting operations, basic/nonbasic variables 23. Simplex algorithm #2 • Correctness (optimality, termination) • Initial basic feasible solution • Anticycling rules (Bland's rule, lexicographic rule) • Worst-case vs. average-case running time 24. LP duality • Primal LP & dual LP • Strong LP duality from simplex algorithm • Separating hyperplane theorem, Farkas' lemma • Strong LP duality from Farkas' lemma	
핵심역량	problem-solving skills, logical thinking	
제10주 4.27.~5.3.	25. Interior-point method & ellipsoid method for LP (and beyond) • Logarithm barrier function, central path, Lagrange multipliers • Path-following methods, step length parameter, convergence analysis • Reduced KKT system, normal equations • Optimization to feasibility, geometric divide-and-conquer, shrinking lemma • Separation oracle, Grötschel-Lovász-Schrijver theorem 26. Heuristics for ILP (Integer Linear Programming) • Gomory's cutting plane method • Branch-and-cut heuristic for (mixed) ILP 27. Gradient descent (for convex optimization) • Unconstrained gradient descent • Constrained gradient descent	
핵심역량	problem discovery and definition, creativity	
제11주 5.4.~5.10.	28. Subproblem analysis #1 • Small vertex covers (fixed-parameter tractability) • Vertex cover in bipartite graph • König-Egerváry theorem (from max-flow/min-cut theorem) 29. Subproblem analysis #2 • Independent set on trees (greedy) • Weighted independent set on trees (DP) • Circular arc coloring based on interval partitioning (fixed-parameter tractability)	5.5.(화) 어린이날
핵심역량	problem discovery and definition, creativity	
제12주 5.11.~5.17.	30. Greedy approximation algorithm • List scheduling for load balancing • Center selection 31. Pricing method & LP rounding • Pricing method for weighted vertex cover • LP rounding for weighted vertex cover • LP rounding for generalized load balancing 32. Fully polynomial-time approximation scheme (FPTAS) • FPTAS for knapsack (rounding) • FPTAS for subset sum (trimming)	

핵심역량	problem discovery and definition, creativity	
제13주 5.18.~5.24.	33. Randomized algorithms #1 • Monte Carlo, RP (Randomized Poly-time) • Las Vegas, ZPP (Zero-error Probabilistic Poly-time) • Contention resolution (Monte Carlo) • Karger's contraction algorithm for global min cut (Monte Carlo) • MAX 3-SAT with linearity of expectation (Las Vegas) 34. Randomized algorithms #2 • Universal hashing, cryptographic hashing • Load balancing (Chernoff bounds)	
핵심역량	problem discovery and definition, creativity	
제14주 5.25.~5.31.	35. Online algorithms #1 • Competitiveness, strong competitiveness • Elevator problem • LRU for fixed-space online caching 36. Online algorithms #2 • VMIN for variable-space online caching • Randomized online caching algorithm	5.25.(월) 대체공휴일
핵심역량	problem-solving skills, logical thinking	
제15주 6.1.~6.7.	37. Quantum computation #1 • Qubits, superposition, measurement (entanglement) • Quantum register, unitary matrices, quantum operations (quantum gates) • BQP, quantum vs probabilistic computation 38. Quantum computation #2 • Quantum circuits, boolean circuits as quantum circuits • Grover's algorithm • Simon's algorithm 39. Quantum algorithms for factoring & discrete logarithm • QFT (Quantum Fourier Transform) • Quantum circuits for modular exponentiation • Shor's order-finding algorithm • Reducing factoring & discrete logarithm to order finding 40. Lattice-based public-key encryption schemes • LWE (Learning With Errors) conjecture • LWE-based public-key encryption scheme • Module-LWE conjecture • Kyber (ML-KEM, Module-Lattice-Based Key-Encapsulation Mechanism) • Lattice problems, worst-case to average-case reduction	6.3.(화) 지방선거일
핵심역량	problem-solving skills, logical thinking	
제16주 6.8.~6.14.	보강 주간	
제17주 6.15.~6.19.	기말고사 (6.18~6.19.)	